1 PERSONAL INFORMATION

Name: Kyle Yilin Gao, PhD
Emails: kyle.gao@aalto.fi, ylg23@cam.ac.uk
Google Scholar: https://scholar.google.ca/citations?user=F-NPRk0AAAAJ&hl=en
ORCID: 0000-0002-8320-6308

Keywords: Computer Vision, Remote Sensing, Photogrammetry, GeoAI, Foundation Models, Embeddings 3D Point Clouds, 3D Reconstruction, LLMs in Geographic Information Systems (GIS)

1.1 Affiliations

Aalto University, Department of Built Environment — Espoo, Finland
Tenure Track Assistant Professor — August 2026 — Ongoing

University of Cambridge, Department of Computer Science and Technology — Cambridge, United Kingdom
Visiting Scholar — July 2026 — May 2028

1.2 Education

University of Waterloo, Waterloo, ON, Canada
Postdoctoral Fellow: Systems Design Engineering — May 2025 — July 2026
Doctor of Philosophy (PhD): Systems Design Engineering. — April 2021 — May 2025

University of Victoria, Victoria, BC, Canada — September 2017 — July 2020
Master of Science (MSc): Physics

University of Waterloo, Waterloo, ON, Canada — September 2011 — July 2016
Honours Bachelor's degree in Mathematics with Distinction

2 AWARDS AND GRANTS

2.1 Project Funding and Scholarships (Postdoctoral)

- NSERC CPRA Fellowship **(Received)** — $70,000 per year
  Natural Sciences and Engineering Research Council of Canada — 2026–2028
  Canada's premier postdoctoral fellowship, consolidating the former NSERC PDF and Banting programs; **ranked 12th out of 187 applicants** (Computer Science committee), qualifying for the previous **Banting Fellowship**. (Received)
- Schmidt AI in Science Postdoctoral Fellowship **(Awarded: Refused for Tenure Track Position)** $95,000 per year
  Eric and Wendy Schmidt Foundation — 2026–2028
  Extremely selective international fellowship spanning nine leading institutions (e.g., Oxford, Cornell, University of Toronto).

2.2 Personal Awards, Scholarships and Funding (PhD Student)

- Ontario Graduate Scholarship (OGS) — $15,000 per year
  Province of Ontario — 2023-2025
- President's Graduate Scholarship — $5,000 per year
  University of Waterloo — 2023-2025

- Mitacs Accelerate Grant (proposal author and project lead) — $30,000 per year
  Mitacs — 2022-2023
- Engineering Dean's Domestic Student Award — $8,500 per year
  University of Waterloo — 2022-2023

3 PUBLICATIONS

3.1 Summary

- **Total citations: 1500+. Peer-reviewed publications: 51**
- 37 peer-reviewed journal articles (32 in Q1 Journals), 16 peer-reviewed conference articles.
- 10 first-authored peer-reviewed papers.

Articles are listed in reverse chronological order. My authorship position is highlighted in **bold**. First-authored works are marked in blue.

3.2 Peer-Reviewed Journal Papers

Peer-reviewed journal publications include 32 Q1 journal papers.

- 15 articles in *International Journal of Applied Earth Observation and Geoinformation*. (Q1, IF = 8.6)
- 5 articles in *IEEE Transactions on Geoscience and Remote Sensing* (Q1, IF = 8.6)
- 1 article in *IPSRS Journal of Photogrammetry and Remote Sensing* (Q1, IF = 12.2)
- 1 article in *Computational Visual Media* (Q1, IF = 18.6)
- 1 article in *IEEE Transactions on Circuits and Systems for Video Technology* (Q1, IF = 11.1)
- 1 article in *IEEE Transactions on Intelligent Transportation Systems* (Q1, IF = 8.4)
- 1 article in *IEEE Transactions on Instrumentation and Measurement* (Q1, IF = 5.9)
- 1 article in *International Journal of Digital Earth* (Q1, IF = 4.9)
- 1 article in *Remote Sensing Applications: Society and Environment* (Q1, IF = 4.5)
- 2 article in *IEEE Geoscience and Remote sensing Letters* (Q1, IF = 4.5)
- 3 article in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (Q1, IF = 6.3)
- 5 articles in quartile $\leq$ Q2 journals.

J36 Q. Guo, L. Han, Q. Xie, L. Li, H. He, and **Gao, Kyle**, "Nu-tsc: Weakly supervised forest/non-forest mapping using public land cover products," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2026

J35 N. Nan, L. Li, L. Han, Z. Liu, **Gao, Kyle**, and H. He, "SGRA: A Spectro-Geometric Rectification Dual-Attention Transformer for Cross-Domain Building Change Detection," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, pp. 1–15, 2026

J34 **Gao, Kyle**, P. Kotta, L. L. Xu, J. Li, and D. A. Clausi, "Securing Multi-Agent GIS Systems: Risk Evaluation and Prompt Hardening Optimization," *IEEE Geoscience and Remote Sensing Letters*, 2026. Accepted

J33 K. Gao, Y. Gao, H. He, D. Lu, L. Xu, and J. Li, "Neural radiance fields in 3d vision: A comprehensive review," *Computational Visual Media*, pp. 1–55, 2026

J32 Y. Jia, Q. Liu, H. He, C. Song, **Gao, Kyle**, Z. Wu, and Y. Chen, "High-Accuracy Water Level Estimation Using Super-Resolution Sentinel-1 Data," *International Journal of Applied Earth Observation and Geoinformation*, vol. 146, p. 105156, 2026

J31 S. Zheng, C. Ye, Y. Li, Q. Xie, Y. Zhou, J. Deng, **Gao, Kyle**, and M. A. Chapman, "SaLSE-net: a self-attention-based local structure enhancement network for landslide point cloud segmentation," *International Journal of Digital Earth*, vol. 19, no. 1, p. 2595413, 2026

J30 **Gao, Kyle**, D. Lu, H. He, L. Li, L. Xu, M. A. Chapman, and J. Li, "Gaussian Building Mesh (GBM): Extract a Building's 3D Mesh with Google Earth and Gaussian Splatting," *Remote Sensing Applications: Society and Environment*, p. 101807, 2025

J29 **Gao, Kyle**, D. Lu, H. He, L. Li, L. Xu, and J. Li, "Digital buildings analysis: 3d modeling, gis integration, and visual descriptions using gaussian splatting, chatgpt/deepseek, and google maps platform," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, 2025. doi: 10.1109/LGRS.2025.3606333

J28 C. Ziyi, L. Tingyu, L. Dilong, G. Jin, W. Cheng, **Gao, Kyle**, L. Jonathan, and G. Zheng, "Leveraging multi-view images to learn domain-invariant discriminative embeddings for cross-view geolocalization," *IEEE Transactions on Circuits and Systems for Video Technology*, 2025. doi:10.1109/TCSVT.2025.3613262

J27 D. Lu, **Gao, Kyle**, J. Li, D. Zhang, and L. Xu, "Exploring token serialization for mamba-based lidar point cloud segmentation," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, 2025. doi:10.1109/TGRS.2025.3605383

J26 **Gao, Kyle**, D. Lu, H. He, L. Li, N. Chen, L. Xu, and J. Li, "Instructor-Worker Large Language Model System for Policy Recommendation: a Case Study on Air Quality Analysis of the January 2025 Los Angeles Wildfires," *International Journal of Applied Earth Observation and Geoinformation*, vol. 143, p. 104774, 2025

J25 D. Lu, J. Zhou, **Gao, Kyle**, L. Xu, and J. Li, "3DLST: 3D Learnable Supertoken Transformer for LiDAR Point Cloud Scene Segmentation," *International Journal of Applied Earth Observation and Geoinformation*, vol. 140, p. 104572, 2025

J24 J. Du, L. Xu, L. Ma, **Gao, Kyle**, J. Zelek, and J. Li, "3D semantic segmentation: Cluster-based sampling and proximity hashing for novel class discovery," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 223, pp. 274–295, 2025

J23 **Gao, Kyle**, D. Lu, H. He, L. Xu, and J. Li, "Enhanced 3D Urban Scene Reconstruction and Point Cloud Densification using Gaussian Splatting and Google Earth Imagery," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, 2025. doi:10.1109/TGRS.2025.3536169

J22 D. Lu, L. Xu, J. Zhou, **Gao, Kyle**, Z. Gong, D. Zhang, and J. Li, "3D-UMamba: 3D U-Net with State Space Model for LiDAR Point Cloud Segmentation," *International Journal of Applied Earth Observation and Geoinformation*, vol. 136, p. 104401, 2025

J21 Z. Chen, H. Wang, X. Wu, J. Wang, X. Lin, C. Wang, **Gao, Kyle**, M. Chapman, and D. Li, "Object detection in aerial images using DOTA dataset: A survey," *International Journal of Applied Earth Observation and Geoinformation*, vol. 134, p. 104208, 2024

J20 N. Yang, L. Li, L. Han, **Gao, Kyle**, S. Qu, and J. Li, "Retrieving heavy metal concentrations in urban soil using satellite hyperspectral imagery," *International Journal of Applied Earth Observation and Geoinformation*, vol. 132, p. 104079, 2024

J19 D. Lu, **Gao, Kyle**, Q. Xie, L. Xu, and J. Li, "3DGTN: 3-D dual-attention GLocal transformer network for point cloud classification and segmentation," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–13, 2024

J18 D. Lu, J. Zhou, **Gao, Kyle**, J. Du, L. Xu, and J. Li, "Dynamic clustering transformer network for point cloud segmentation," *International Journal of Applied Earth Observation and Geoinformation*, vol. 128, p. 103791, 2024

J17 Y. Cai, B. Hu, H. He, **Gao, Kyle**, H. Xu, Y. Zhang, S. Pirasteh, X. Wang, W. Chen, and H. Li, "Automatic error correction: Improving annotation quality for model optimization in oil-exploration related land disturbances mapping," *The Egyptian Journal of Remote Sensing and Space Sciences*, vol. 27, no. 1, pp. 108–119, 2024

J16 Z. Gong, R. He, **Gao, Kyle**, and G. Cai, "Scene-aware Online Calibration of LiDAR and Cameras for Driving Systems," *IEEE Transactions on Instrumentation and Measurement*, p. 1–1, 2023

J15 L. Li, L. Han, **Gao, Kyle**, H. He, L. Wang, and J. Li, "Coarse-to-fine matching via cross fusion of satellite images," *International Journal of Applied Earth Observation and Geoinformation*, vol. 125, p. 103574, 2023

J14 L. Li, L. Han, M. Liu, **Gao, Kyle**, H. He, L. Wang, and J. Li, "SAR-Optical Image Matching with

Semantic Position Probability Distribution," *IEEE Transactions on Geoscience and Remote Sensing*, 2023

J13 Z. Chen, Y. Luo, Y. Chen, J. Wang, D. Li, **Gao, Kyle**, C. Wang, and J. Li, "BrGAN: Blur Resist Generative Adversarial Network With Multiple Joint Dilated Residual Convolutions for Chlorophyll Color Image Restoration," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–12, 2023

J12 **Gao, Kyle**, H. He, D. Lu, L. Xu, L. Ma, and J. Li, "Optimizing and Evaluating Swin Transformer for Aircraft Classification: Analysis and Generalizability of the MTARSI Dataset," *IEEE Access*, vol. 10, pp. 134427–134439, 2022

J11 X. Lei, H. Guan, L. Ma, Y. Yu, Z. Dong, **Gao, Kyle**, M. R. Delavar, and J. Li, "WSPointNet: A multi-branch weakly supervised learning network for semantic segmentation of large-scale mobile laser scanning point clouds," *International Journal of Applied Earth Observation and Geoinformation*, vol. 115, p. 103129, 2022

J10 D. Lu, Q. Xie, **Gao, Kyle**, L. Xu, and J. Li, "3DCTN: 3D convolution-transformer network for point cloud classification," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 12, pp. 24854–24865, 2022

J9 W. Liu, J. Liu, Z. Luo, H. Zhang, **Gao, Kyle**, and J. Li, "Weakly supervised high spatial resolution land cover mapping based on self-training with weighted pseudo-labels," *International Journal of Applied Earth Observation and Geoinformation*, vol. 112, p. 102931

J8 Y. Lin, T. Zhang, X. Liu, J. Yu, J. Li, and **Gao, Kyle**, "Dynamic monitoring and modeling of the growth-poverty-inequality trilemma in the Nile River Basin with consistent night-time data (2000–2020)," *International Journal of Applied Earth Observation and Geoinformation*, vol. 112, p. 102903, 2022

J7 H. He, H. Xu, Y. Zhang, **Gao, Kyle**, H. Li, L. Ma, and J. Li, "Mask R-CNN based automated identification and extraction of oil well sites," *International Journal of Applied Earth Observation and Geoinformation*, vol. 112, p. 102875, 2022

J6 H. He, **Gao, Kyle**, W. Tan, L. Wang, S. N. Fatholahi, N. Chen, M. A. Chapman, and J. Li, "Impact of Deep Learning-Based Super-Resolution on Building Footprint Extraction," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 43, pp. 31–37, 2022

J5 H. He, **Gao, Kyle**, W. Tan, L. Wang, N. Chen, L. Ma, and J. Li, "Super-resolving and composing building dataset using a momentum spatial-channel attention residual feature aggregation network," *International Journal of Applied Earth Observation and Geoinformation*, vol. 111, p. 102826, 2022

J4 **Gao, Kyle**, M. Chen, S. Narges Fatholahi, H. He, H. Xu, J. Marcato Junior, W. Nunes Gonçalves, M. A. Chapman, and J. Li, "A region-based deep learning approach to instance segmentation of aerial orthoimagery for building rooftop extraction," *Geomatica*, vol. 75, no. 1, pp. 148–164, 2022

J3 H. He, Z. Jiang, **Gao, Kyle**, S. Narges Fatholahi, W. Tan, B. Hu, H. Xu, M. A. Chapman, and J. Li, "Waterloo building dataset: A city-scale vector building dataset for mapping building footprints using aerial orthoimagery," *Geomatica*, vol. 75, no. 3, pp. 99–115, 2022

J2 P. Zhao, H. Guan, D. Li, Y. Yu, H. Wang, **Gao, Kyle**, J. M. Junior, and J. Li, "Airborne multispectral LiDAR point cloud classification with a feature Reasoning-based graph convolution network," *International Journal of Applied Earth Observation and Geoinformation*, vol. 105, p. 102634, 2021

J1 N. Chen, L. Sui, B. Zhang, H. He, **Gao, Kyle**, Y. Li, J. M. Junior, and J. Li, "Fusion of Hyperspectral-Multispectral images joining Spatial-Spectral Dual-Dictionary and structured sparse Low-rank representation," *International Journal of Applied Earth Observation and Geoinformation*, vol. 104, p. 102570, 2021

3.3 Peer-Reviewed Conference Papers

C16 Q. Wu, **Gao, Kyle**, D. Long, D. A. Clausi, J. Li, and Y. Chen, "KitchenTwin: Semantically and Geometrically Grounded 3D Kitchen Digital Twins," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR-W)*, 2026

C15 **Gao, Kyle**, J. Cumming, J. Li, L. Xu, and D. A. Clausi, "Risk-Aware LLM Agents for Geospatial Data Retrieval: Design and Preliminary Adversarial Evaluation," in *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, ISPRS, 2026

C14 N. Azad, J. Noa Turnes, **Gao, Kyle**, H. He, L. Xu, and D. A. Clausi, "Benchmarking input-level sentinel-1 and radarsat constellation mission fusion for multi-task sea ice mapping on an extended ai4arctic dataset," in *Proceedings of the IEEE OCEANS Sanya Conference*, IEEE, 2026

C13 Q. Wu, **Gao, Kyle**, W. Sun, H. He, Y. Chen, D. A. Clausi, and J. Li, "Monocular metric 3d reconstruction for remote island surveying via geometry-consistent transformer and orbital imagery," in *Proceedings of the IEEE OCEANS Sanya Conference*, IEEE, 2026

C12 D. Szczecina, N. Azad, H. Sun, **Gao, Kyle**, A. Bertnyk, and L. L. Xu, "Sparse Data Tree Canopy Segmentation: Fine-Tuning Leading Pretrained Models on Only 150 Images," in *Proceedings of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*, 2026

C11 Q. Wu, **Gao, Kyle**, W. Sun, Z. Xu, H. Sun, L. Xu, Y. Chen, D. A. Clausi, and J. Li, "Rapid Forest Fuel Load Estimation via Virtual Remote Sensing and Metric-Scale Feed-Forward 3D Reconstruction," in *Proceedings of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*, 2026

C10 H. Sun, **Gao, Kyle**, Q. Wu, N. Azad, L. L. Xu, and D. A. Clausi, "Label-Free Forest–Urban Segmentation Using AlphaEarth Foundations," in *Proceedings of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*, 2026

C9 Q. Wu, **Gao, Kyle**, W. Sun, Y. Chen, D. A. Clausi, and J. Li, "Real-Scale Island Area and Coastline Estimation using Only its Place Name or Coordinates," in *Proceedings of the OCEANS Conference & Exposition*, 2026. Submitted

C8 X. Huang, Z. Xu, H. Wu, J. Wang, Q. Xia, Y. Xia, J. Li, **Gao, Kyle**, C. Wen, and C. Wang, "L4DR: LiDAR-4DRadar Fusion for Weather-Robust 3D Object Detection," in *AAAI Conference on Artificial Intelligence (AAAI-25)*, 2025

C7 N. Chen, B. Zhang, H. He, Z. Liu, and **Gao, Kyle**, "Synchronizing Spatiotemporal Reflectance Fusion via Dual Bayesian Nonparametric Inference and Explicit Downsampling," in *IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium*, pp. 10358–10362, IEEE, 2024

C6 D. Chen, J. Xiao, **Gao, Kyle**, Y. Lu, S. Fatholahi, and J. Li, "Natural Language Aided Remote Sensing Image Few-Shot Classification," in *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*, pp. 6298–6301, IEEE, 2023

C5 X. Liu, S. Qiao, **Gao, Kyle**, H. He, L. Ma, and J. Li, "Nighttime Light Missing Data Retrieval Using Modis Version 6 Satellite Data and Mask Dilated Partial Convolutional Neural Network," in *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*, pp. 2989–2992, IEEE, 2023

C4 D. Chen, T. Cheng, Y. Lu, **Gao, Kyle**, S. Fatholahi, and J. Li, "Research on Fast Detection Method of Wind Turbine in Remote Sensing Image Land Area Based on YOLO," in *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium*, pp. 2823–2826, IEEE, 2023

C3 **Gao, Kyle** and S. R. Koscielniak, "Beam-Beam Resonance Widths in the HL-LHC, and Reduction by Phasing of Interaction Points," in *Proc. 13th International Particle Accelerator Conference (IPAC'22)*, pp. 2280–2283, 2022

C2 H. He, K. Yang, Y. Cai, and **others**, "The impact of data volume on performance of deep learning based building rooftop extraction using very high spatial resolution aerial images," in *2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS*, pp. 1343–1346, IEEE, 2021

C1 H. He, Z. Jiang, W. Tan, Y. Cai, S. N. Fatholahi, **Gao, Kyle**, H. Xu, B. Hu, L. Qing, and J. Li, "Waterloo Building Dataset: A large-scale very-high-spatial-resolution image dataset for building rooftop extraction," *Abstracts of the ICA*, vol. 3, pp. 1–2, 2021

3.4 Conference Presentations

- *IGARSS 2026* August 2026
  Title: "Rapid Forest Fuel Load Estimation via Virtual Remote Sensing and Metric-Scale Feed-Forward 3D Reconstruction"
- *IGARSS 2026* August 2026
  Title: "Tree Canopy Segmentation in Low-Data Regimes Using Pretrained Deep Models"
- *ISPRS Congress 2026, CSRS: 47th Canadian Symposium on Remote Sensing* July 2026
  Title: "VGGT-SLAM for 3D Reconstruction of Low-Altitude Remote Sensing Data: Feasibility and Limitations"
  ISPRS Congress 2026, CSRS: 47th Canadian Symposium on Remote Sensing July 2026
  Title: "Interactive 3D Scene Segmentation for Construction Sites via Gaussian Splatting and Foundation Models"
- *ISPRS Congress 2026, CSRS: 47th Canadian Symposium on Remote Sensing* July 2026
  Title: "Digital Building Analysis (DBA): Cloud-GIS-Based 3D Building Modelling and Multi-Agent AI Analytics Using Gaussian Splatting and Google Maps Platform"
- *ISPRS Congress 2026, CSRS: 47th Canadian Symposium on Remote Sensing* July 2026
  Title: "Advances in 3D Urban Reconstruction and Building Mesh Extraction using Gaussian Splatting and Google Earth"
- *CVPR 2026: Workshop on Foundation and Large Vision Models in Remote Sensing* June 2026
  Title: "BuildingTwin: Geometrically Grounded Building-Centric 3D Reconstruction for Urban Digital Twins"

3.5 Select Preprints Under Peer-Review (PR) and Others (O)

PR2 K. Xiong, S. Xu, **Gao, Kyle**, S. Ao, B. Yang, S. Shen, C. Wen, and C. Wang, "INTEGER++: Scalable Unsupervised Point Cloud Registration via Mining and Transferring Feature-Geometry Coherence," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026. Submitted, second review round

PR1 J. N. Turnes, M. Patel, F. J. Pena Cantu, **Gao, Kyle**, L. Xu, and D. A. Clausi, "Is Pre-training Enough? SAR-IceFM: Towards Multi-Task Foundation Models for Sea Ice Classification." Submitted to *Remote Sensing of Environment*, 2026

O2 **Gao, Kyle**, "Towards Urban Digital Twins With Gaussian Splatting, Large-Language-Models, and Cloud Mapping Services," *PhD Thesis*, 2025

O1 **Gao, Kyle**, "Beam-Beam Resonance Widths in the HL-LHC, and Reduction by Phasing of Interaction Points," *Msc Thesis*, 2020

4 TEACHING EXPERIENCE

Supervision and Mentorship

Teaching

University of Waterloo, Department of Geography and Environmental Management
Sessional Lecturer: Geography 316 (Multivariate Statistics in R) September 2024 — December 2024

University of Waterloo, Department of Systems Design Engineering
Sessional Lecturer: SYDE 572 (Introduction to Pattern Recognition) January 2024 — April 2024

University of Waterloo, Department of Geography and Environmental Management Waterloo, Canada
Teaching Assistant and Guest Lecturer-6 courses September 2021 — December 2023

University of Victoria, Department of Physics and Astronomy Victoria, Canada
Teaching Assistant and Lab Assistant-2 courses September 2017 — September 2018

5 Research Employment

Aalto University, Department of Built Environment Espoo, Finland
Tenure Track Assistant Professor of Geoinformatics August 2026 – Present

- Leading research group in GeoAI, 3D computer vision, and remote sensing for environmental resilience and urban digital twin research.
- Supervising graduate students and developing international research collaborations.

University of Cambridge, Department of Computer Science and Technology Cambridge, United Kingdom
Visiting Scholar July 2026 – May 2028

- Planning international collaborations on GeoAI and earth foundation models, 3D reconstruction for digital twins.
- Preparing consortium grant applications on geoAI and remote sensing for environmental monitoring, disaster management, and climate studies.

University of Waterloo, GIM Lab, VIP Lab Waterloo, Canada
Postdoctoral Fellow May 2025 –

- Directing laboratory research on urban 3D reconstruction using Gaussian Splatting.
- Developing climate risk assessment frameworks via cloud-based geographic information systems.
- Advancing self-supervised foundation models for remote sensing and SAR-based sea-ice analysis.
- Managing preparation of multiple manuscripts across these core research themes.
- Awarded prestigious Schmidt AI Fellowship, NSERC CPRA Fellowship.

University of Waterloo, Geospatial Intelligence and Mapping Lab Waterloo, Canada
Graduate Research Assistant September 2020 – April 2025

- Executed diverse research initiatives producing over 30 peer-reviewed publications.
- Obtained 60,000 $ in Mitacs funding and 50,000 $ in academic scholarships.
- Aided in securing a 310,000 $ NSERC Discovery Grant alongside the primary investigator (This grant is not included in my personal CV).

TRIUMF, Accelerator Physics Group Vancouver, Canada
Graduate Research Assistant-Particle Accelerator Physics January 2018 – July 2020

- Modeled electromagnetic fields within the Large Hadron Collider using Lie algebra.

Undergraduate Research Assistant (Co-op internships) Multiple Positions Canada
Undergraduate Research Assistant

- *National Research Council of Canada, Quantum Physics Group*: Demonstrated quantum Hall effects in low-temperature experiments on electron dot quantum computing chips.
- *TRIUMF, Accelerator Physics Group*: Created mathematical model of linear electron accelerator.
- *Health Canada*: Parallelized Monte Carlo simulations of radiation on the human body.
- *University of Waterloo, Department of Physics and Astronomy*: Created visualization software and web interface for galactic Doppler shift data (2M++).

6 PROFESSIONAL AND ACADEMIC SERVICES/ACTIVITIES

6.1 Conferences Organization/Committee

- **Chair of Community Contributed Theme**, *2026 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*: "Advances in GeoAI Systems for Wildfire Monitoring".
- **Chair of Community Contributed Theme**, *2026 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*: "Vision-Language Models for Remote Sensing: Foundations, Applications, and Challenges" (3 sessions).
- **Chair of Community Contributed Theme**, *2026 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*: "Scalable and Efficient Processing of Large-Scale LiDAR Point Clouds".
- **Chair of Community Contributed Theme**, *2026 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*: "Vision-Language Models for Remote Sensing: Foundations, Applications, and Challenges".
- **Chair of Community Contributed Theme**, *2026 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*: "Remote Sensing and Geospatial Modeling of Wildfire Risk, Detection, and Recovery".
- **Chair of Thematic Session**, *XXV Congress of the International Society for Photogrammetry and Remote Sensing (ISPRS Congress 2026)*: "Multimodal Large Language Models for Remote Sensing".
- **Co-Chair of Invited Session**, *XXV Congress of the International Society for Photogrammetry and Remote Sensing (ISPRS Congress 2026)*: "Canadian Remote Sensing for Urban Applications".
- **Co-chair, Main Organizing Committee** of *11th Annual Conference on Vision and Intelligent Systems (CVIS 25)*.
- **Program Committee** for Main Track and AI Alignment Track: *40th Annual AAAI Conference on Artificial Intelligence (AAAI 26)*.

6.2 Journal Editorial Board

- **Article Collection Editor** *GIScience and Remote Sensing*: "Multimodal Remote Sensing Data Fusion for Geospatial Intelligence."
- **Special Issue Editor** *Photogrammetric Engineering and Remote Sensing*: "Special Issue on Large Language Model in Remote Sensing: Across Different Modalities."
- **Special Issue Editor** *Electronics*: "Special Issue on Multimodal Remote Sensing for Wildfire Monitoring."

6.3 Invited Speaker

- *Aalto University* May 2026
  Title: "Extending classical photogrammetry with learning systems; towards digital twins"
- *Chang'an University* July 2025
  Title: "An overview of the applications of Large Language Models to Remote Sensing and Geographic Information Systems"
- *East China Normal University* July 2025
  Title: "Lie Algebra in Computer Vision, from Homography to SLAM"

6.4 Peer-Reviewer

I have peer-reviewed over 94 manuscripts for the following journals.

- *ISPRS Journal of Photogrammetry and Remote Sensing*
- *Remote Sensing of Environment*
- *IEEE Transactions on Pattern Analysis and Machine Intelligence*

- *IEEE Transactions on Geoscience and Remote Sensing*
- *IEEE Transactions on Neural Networks and Learning Systems*
- *IEEE Transactions on Visualization and Computer Graphics*
- *IEEE Geoscience and Remote Sensing Letters*
- *IEEE Transactions on Intelligent Transportation Systems*
- *International Journal of Applied Earth Observation and Geoinformation*
- *ACM Transactions on Multimedia Computing, Communications and Applications*
- *Neurocomputing*

6.5 Professional Memberships and Leadership Positions

Co-Chair 2026 — Ongoing
TCIII/II Working Group on Foundations of AI/ML for Photogrammetry and Remote Sensing
International Society for Photogrammetry and Remote Sensing (ISPRS)

Professional Member 2024 — Ongoing
Association for Computing Machinery

Member 2023 — Ongoing
Geoscience and Remote Sensing Society (GRSS)

Member 2021 — Ongoing
Institute of Electrical and Electronics Engineers (IEEE)

Student Representative of Canada on the ICA 2023 — 2025
International Cartographic Association (ICA)
Commission on Geospatial Data Analytics

Student Committee Vice Chair 2023 — 2025
Canadian Institute of Geomatics (CIG)

Waterloo Chapter Student Representative 2020 — 2025
Canadian Remote Sensing Society (CRSS)